



# Grand Rounds Health Expert Opinion

## About Dr. Anuj Patel

Dr. Anuj Patel is a board-certified medical oncologist at Dana-Farber Cancer Institute. He earned his medical degree from The University of Pittsburgh School of Medicine, and continued his medical training with a residency in internal medicine from The University of Pittsburgh Medical Center. Following his residency, Dr. Patel completed his medical training with a fellowship in hematology/oncology from The University of Pittsburgh Medical Center. Dr. Patel specializes in gastrointestinal cancers. He is currently an Instructor of Medicine at Harvard Medical School as well as Attending Physician at Dana-Farber Cancer Institute.

### Institutional Affiliation

Instructor of Medicine, Harvard Medical School; Medical Oncologist, Dana-Farber Cancer Institute

### Education

University of Pittsburgh School of Medicine

### Residency

University of Pittsburgh Medical Center

### Fellowship

University of Pittsburgh Medical Center

### Specialties

Biliary cancer, colon cancer, anal cancer, esophageal cancer, gallbladder cancer, gastrointestinal carcinoid tumor, liver cancer, neuroendocrine/carcinoid tumors, pancreatic cancer, rectal cancer, stomach cancer

**Publications** <https://www.ncbi.nlm.nih.gov/pubmed/?term=anuj+patel>

## Summary of Patient Medical History

You were recently diagnosed with pancreatic cancer. You underwent tests and procedures for weight loss and diarrhea last year; these eventually led to CT scans showing a tumor in your pancreas. This tumor was biopsied and found to be a form of pancreatic

cancer. Your scans also showed concerning lesions in your lungs; one of these lesions was biopsied as well.

You have been recommended treatment with chemotherapy. The goal of your Expert Medical Opinion is to better understand your diagnosis and determine the best steps moving forward.

## **Background Information**

I'd like to provide you with a bit of background information about pancreatic cancer. Although you may be familiar with some of what I'm going to tell you, I feel that the information will be helpful in better understanding the recommendations that I make.

### **Pancreatic cancer**

Pancreatic cancer forms within the pancreas, an organ that lies behind the lower part of the stomach. Normally, the pancreas secretes enzymes into the intestines which help to break down food. It also secretes hormones that help regulate sugar metabolism. Pancreatic adenocarcinoma (also known as pancreatic ductal adenocarcinoma) is the most common type of pancreatic cancer. It arises from the ductal cells in the pancreas, which help form the digestive component of the pancreas.

Pancreatic adenocarcinoma is a cancer that is being diagnosed more frequently in the United States. For earlier-stage pancreatic cancers, surgery can be sometimes be considered. It is not always possible to remove even smaller cancers surgically, however, as the pancreas is adjacent to a number of major organs and blood vessels. Even if the cancer is able to be removed, it comes back frequently after surgery. Because of this chemotherapy and, sometimes, radiation are often also recommended before or after surgery. Unfortunately, even with all of these measures, pancreatic cancer recurs in a very high percentage of patients.

The majority of patients will present with advanced disease that is not potentially curable through surgery. In some patients, this is because the cancer is too closely involved with those blood vessels near the pancreas; in others, it is because their cancer has already spread to other parts of the body. For all of these patients, chemotherapy is the mainstay of treatment. The goals of chemotherapy in this advanced setting are mainly to control the disease; however, chemotherapy alone is not able to cure pancreatic cancer.

## **Summary of Opinion**

Your scans and the ultrasound performed during your December endoscopy show a mass in the front of your pancreas, involving the regions known as the pancreatic head and uncinate process. This mass was biopsied and, after review here, was found to represent pancreatic cancer - specifically the primary form of pancreatic cancer, known as pancreatic adenocarcinoma.

Your scan show that this tumor extends along a number of the major blood vessels adjacent to the pancreas. Most importantly, it surrounds one of the critical arteries here, called the superior mesenteric artery (SMA). This artery cannot be removed or reconstructed during surgery - pancreatic surgeons need to be able to separate the tumor from these major arteries safely; any involvement of the tumor with these arteries increases the likelihood that some amount of cancer will be left behind during an operation. When tumors wrap around these arteries more than 180°, we know that surgical resection will not be possible in a safe and clean fashion. With these findings, your primary tumor would be classified as locally-advanced or unresectable.

Your scans also showed several nodules in your lungs; these nodules had features concerning for cancer - but could also have represented a different active process, such as an infection. You underwent a biopsy of one of these lung nodules last month. The pathology report shows that cancer was found in this biopsy. This cancer was also an adenocarcinoma - the same form of cancer found in your pancreatic biopsy. Additional tests showed that it has features most commonly seen in cancers that originate in the GI tract, rather than in the lungs. There are rare lung cancers that can have these GI features; however, with all of your information available, I do think these findings are more consistent with these nodules being metastatic sites from your pancreatic cancer.

Your pathologists have ordered additional tests to help confirm this; I think it will be helpful to follow up on those test results, when available. <<Christa: They were available on 1-12 and re-reviewed on 1/19. Unsure why these results were not shared by Grand Rounds for this review.>> It would also be reasonable to have this biopsy reviewed at Brigham & Women's Hospital/Dana-Farber - as they did with your initial pancreatic tumor biopsy.

I agree with your recommended treatment plan for chemotherapy. With both locally advanced pancreatic adenocarcinoma, when surgery is not possible, and metastatic pancreatic adenocarcinoma, the primary treatment is chemotherapy. Pancreatic cancer, particularly metastatic pancreatic cancer, is a more systemwide process - in addition to these sites of cancer that we can see on your scans, there are microscopic sites of cancer that we cannot see. Because of this, the focus of your treatment needs to involve systemwide therapies that attack all of these sites of disease. In pancreatic adenocarcinoma, the only proven systemwide therapies involve chemotherapy.

I also agree with the chemotherapy treatments that your oncologist has discussed. There are two primary chemotherapy regimens used for the first-line treatment of metastatic pancreatic adenocarcinoma. One is your regimen, FOLFIRINOX; <<Christa: This is not her regimen, from January 4, the plan was the gemcitabine and Abraxane. Unclear why the expert is misinformed on this point.>> the other regimen involves the combination of two other chemotherapy drugs: gemcitabine and nab-paclitaxel (Abraxane). Both of these regimens were studied in separate, large, multinational trials of patients with metastatic pancreatic adenocarcinoma and both regimens showed an improvement in the control of disease and survival of patients when compared to the prior standard of care; the two regimens have never been compared head-to-head. With this information, we think of both regimens as appropriate first-line treatments for this cancer.

## Clinical Trials

There are clinical trials for which you might be eligible. Active clinical trials are listed on the website [clinicaltrials.gov](https://clinicaltrials.gov). However, it is often difficult to process the wide range of trials listed on this site. It is also important to note that clinical trials have specific criteria for inclusion/exclusion and while trials may be listed as actively recruiting, they may not currently have any open slots for enrollment. In some cases, you can find listings of trials on the websites of academic centers in your region. It is often more practical, though, to work with your oncologist to explore appropriate clinical trials in your area.

I will note that most of the clinical trials studying therapies for the first line of treatment for patients with pancreatic adenocarcinoma still involve chemotherapy. For example, at our institution (Dana-Farber), we currently have one first-line pancreatic cancer trial open: <https://clinicaltrials.gov/ct2/show/NCT03520790> (<https://clinicaltrials.gov/ct2/show/NCT03520790>)

In this trial, all patients receive the standard chemotherapy regimen gemcitabine/nab-paclitaxel, which I mentioned above. Patients are then randomly assigned to one of three different arms: in one arm, patients receive IV treatment with paricalcitol (a synthetic form of vitamin D); in the second arm, patients receive oral paricalcitol; in the third arm, patients receive a placebo. The goal of this trial is to study whether we can improve outcomes for patients receiving chemotherapy with gemcitabine/nab-paclitaxel - however, the foundation of treatment is still chemotherapy.

There are sometimes trials for later lines of treatment that explore more targeted treatments instead of chemotherapy. These trials are often tailored to specific mutations or genomic findings in specific patients' cancers. I would consider sending the tissue from your biopsies for more in-depth profiling of your cancer to look for these genomic findings, through a process called next-generation sequencing (NGS). NGS evaluates the cancer for a broad range of genes that are known to be altered in different cancers. NGS is often performed to see if a patient's cancer holds any mutations that might be relevant to treatment decisions, such as the choice of specific chemotherapy or enrollment in a clinical trial.

I will note that your pancreatic biopsy was performed using a technique known as fine-needle aspiration. This is often a more straightforward procedure - but it returns only a small sample. It is likely that there is not enough tissue from your original biopsy for NGS; I am more hopeful that there will be enough tissue present in your lung sample. If there is not, it would be reasonable to obtain a repeat biopsy of the pancreas or a lung nodule. Another, less invasive option, however, would be to obtain a "liquid biopsy" - this involves sequencing of cancer DNA identified in your blood, through a blood test. NGS can sometimes be performed through a liquid biopsy, though this may not be as thorough or as high-yield as if it were performed on a true tissue biopsy.

I would also consider another form of genetic testing known as germline testing; unlike NGS, which primarily looks at the genetics/genomics of your cancer, germline testing is performed to see if patients have any inherited risk factors that contributed to their developing cancer. Approximately 10 to 20% of patients with pancreatic cancer have some inherited risk factor that predisposed them to develop pancreatic cancer. We find these even in a significant number of patients without any family history. Because of this, at our institution, we routinely recommend that all patients with a new diagnosis of pancreatic cancer meet with a genetic counselor in a genetics clinic with experience in cancer risk assessment to discuss testing. National guidelines also recommend germline testing for any patient with confirmed pancreatic cancer. If an inherited mutation is found, it can have an impact on your treatment; it can also have implications for children, siblings, and other family members.

The management of cancer is complicated because the state of your cancer, your health, and your goals may change over time. There is rarely a clear right approach for any decision - I think that this is particularly true in your case, as you were only recently diagnosed with this cancer and are still undergoing tests to better understand your cancer and your options for treatment. It will be important to continue to speak with your providers at each step in your care, both about the potential risks and benefits of any treatment option and to continue to express your personal goals of care.

## Recommendations from Dr. Anuj Patel

I have not had the opportunity to examine you and talk to you face-to-face. I am therefore not your treating physician. You should use this consultative second opinion that I am providing to you and your treating physician as a basis of discussion with your treating physician. You need to work with your treating physician, who can examine you in person, to determine the optimal treatment for you.

Grand Rounds does not determine whether a procedure is covered by your health plan. Contact your health plan provider to determine what is covered.

Let me summarize my thoughts and recommendations for you:

- I would review the findings from your lung biopsy with your providers and follow up on new results as they become available.
- I agree with your current plan for treatment with chemotherapy.
- I would continue to explore clinical trials in your region, particularly at any point that you need to make a decision or change for your treatment.
- I would consider sending tissue from your biopsies for next-generation sequencing (NGS).
- If there is not enough available tissue from either biopsy to perform NGS, you could consider a repeat biopsy. An alternative would be to obtain a "liquid biopsy."
- I would recommend evaluation in a genetics clinic with experience in cancer risk assessment.

I hope you have found this information helpful and that it leads to a useful discussion with your providers.

## Links for Patient

Three sites that I have found that provide useful, factual information are:

- American Cancer Society <https://www.cancer.org/cancer/pancreatic-cancer.html>
- NIH Medline Plus <https://medlineplus.gov/pancreaticcancer.html>
- American Society of Clinical Oncology <https://www.cancer.net/cancer-types/pancreatic-cancer>